

Lab #:

Date:

Title: Standardization of sodium hydroxide with a weak acid

Topic: Primary Standards and Standard solutions

Skill(s):

Aim: To standardize a solution of sodium hydroxide using oxalic acid dihydrate.

Apparatus/Materials:

Diagram:

Method:

1. Wash and dry a 250 mL beaker.
2. Place the clean, dry beaker on the top-loading balance and tare the balance with the beaker on it.
3. Accurately weigh 0.94 g of oxalic acid dihydrate in the beaker.
4. Add approximately 200 mL of distilled water to the oxalic acid and stir using a stirring rod until all the solid dissolves.
5. Transfer the solution to a 250 mL volumetric flask with the aid of a funnel. Rinse the beaker and funnel with small portions of distilled water and add rinses to the volumetric flask to avoid loss.
6. Fill the flask to the calibration mark with distilled water at roughly room temperature. Add water dropwise using a dropper pipette as the mark is approached so the bottom of the meniscus sits on the mark.
7. Stopper and gently mix the volumetric flask several times to ensure homogeneity.
8. Label the flask with the relevant concentration and volume of its contents.

9. Secure the burette on a clamp and stand and fill it with sodium hydroxide solution until the sodium hydroxide solution is just below the zero mark.
10. Take the initial reading of the burette.
11. Pipette 25 mL of the oxalic acid into a 250 mL Erlenmeyer flask and add three (3) drops of phenolphthalein indicator to the flask.
12. Titrate the oxalic acid with the sodium hydroxide until a permanent pink colour appears in the flask.
13. Record the final reading of the burette.
14. Repeat steps 11 to 13 until concordant values are obtained.

Result:

Trial	1	2	3
Initial burette reading (mL)			
Final burette reading (mL)			
Volume of NaOH used (mL)			

Calculations:

1. Calculate the number of moles of oxalic acid dihydrate that was weighed.
2. Write the balanced chemical equation for the reaction between oxalic acid and sodium hydroxide.
3. Hence, determine the number of moles of sodium hydroxide that reacted with the oxalic acid dihydrate.
4. Determine the molar concentration of the oxalic acid dihydrate solution that was prepared.
5. Calculate the average titre volume of NaOH that was used.

6. Hence, determine the molar concentration of sodium hydroxide.
7. Calculate the mass concentration of sodium hydroxide.

Discussion:

- Why is it necessary to standardize sodium hydroxide before using it in titrations?
- Why is oxalic acid dihydrate suitable as a primary standard in this experiment?
- Why prepare a solution of oxalic acid of known concentration rather than weighing solid NaOH directly?
- Why is it important to carry out multiple titrations and obtain concordant results?
- In what real-world situations is standardization of NaOH (or other reagents) necessary?

Precautions:

Sources of Error:

Conclusion: